

Skills Summary

Data Displays and Shape Properties

Use this reference while completing your worksheet or when studying.

Skill 1 — Reading a Bar Graph or Line Plot

Match the question to the move:

“How many *are there* of one kind?” $\Rightarrow$ read one bar or count one column.

“How many *altogether*?” $\Rightarrow$ add every bar or column.

“How many *more* than?” $\Rightarrow$ subtract one from the other.

On a line plot the labels are the values and the $\times$ marks are the things counted.

Example: A graph shows red 7, blue 4, green 2. How many more red than blue?

Step 1. The words how many more compare two groups, so this is a subtraction and I only need two of the three bars.

Step 2. Read the two bars and subtract the smaller from the larger: $7 - 4$.
more red than blue = **3**

Try it: Same graph (red 7, blue 4, green 2). How many are there altogether?

check: 13

Skill 2 — Testing a Shape’s Attributes

Test one attribute at a time.

Square corners: the corner of a card fits exactly. A square and a rectangle have four; a rhombus and a trapezoid have none.

Parallel sides: two sides that stay the same distance apart. A square, rhombus and rectangle each have two pairs; a trapezoid has one.

Having four sides is not an attribute that decides anything — every quadrilateral has four.

Example: How many square corners does a rhombus have?

Step 1. The question names one attribute, so I test only the corners and put the side lengths out of my mind.

Step 2. A rhombus is a square pushed over: two of its corners are sharp and two are blunt, so a card corner fits none of them.

square corners on a rhombus = **0**

Try it: How many pairs of parallel sides does a trapezoid have?

check: 1

Skill 3 — Using a Shape Fact Inside a Data Question

Two steps, in this order:

- (1) Use the shape rule to decide which groups belong to the question.
- (2) Only then add or subtract those groups' numbers.

Doing step 2 first is the usual mistake: adding every bar answers “how many altogether,” which is almost never the question being asked.

Example: A box holds 5 squares, 4 rectangles and 2 trapezoids. How many quadrilaterals is that?

Step 1. Before adding anything I check which groups fit the word in the question, because a quadrilateral is any shape with four straight sides.

Step 2. All three kinds have four sides, so all three groups count: $5 + 4 + 2$.
quadrilaterals = **11**

Try it: From the same box (5 squares, 4 rectangles, 2 trapezoids), how many shapes have four square corners?

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(!) Watch out: read the question twice before you add. “How many altogether,” “how many quadrilaterals” and “how many have four square corners” use the same graph and give three different answers.